

AMIRASSYL SAGYNDYK

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EXPERIENCE

Exploratorium

Software Engineer, Civic Project Partnership (via Minerva University)

September 2025 – May 2026

- Built the interactive front end for a museum exhibit prototype in a team of 5, coordinating real-time multimodal inputs across camera, microphone, and dynamic UI elements for a seamless exhibit.
- Integrated ElevenLabs voice synthesis and MediaPipe computer-vision pose tracking into a unified front-end event loop, translating real-time visitor gestures and speech into responsive UI animations.
- Delivered a working exhibit prototype to the Exploratorium for internal review, collaborating with research fellows to wireframe and implement visitor journeys.

Minerva University

Mathematics Research Assistant

September 2025 – May 2026

- Developed and ran Python numerical solvers for Navier–Stokes boundary-value problems, modeling non-wetting droplet rebound behavior across varying viscosity and impact velocity conditions.

xCellec Robotics

Programmer, FIRST Robotics Team

September 2023 – June 2024

- Developed C/C++ control logic and hardware–software interfaces for Arduino-based robotic systems; team earned the FIRST Championship Design Award (Central Asia) and Samsung Solve for Tomorrow Semifinalist placement.
- Produced technical explainer videos demonstrating robot logic and design that won the Social Media Challenge (190K+ views).

Yale University

Yale Young Global Scholars Program Fellow

June 2026 – August 2026

- Mentored 750+ international high-school students across 3 academic sessions; coordinated campus-wide programming, including curation of a competitive speaker series.

EDUCATION

Minerva University

Bachelor of Science, Computational Sciences

Expected May 2029

- Relevant Coursework: Probability and Statistics, Discrete Math, Computational Modeling, Game Theory, Data Structures and Algorithms, Single and Multivariable Calculus, Linear Algebra

Nazarbayev Intellectual School of Physics and Mathematics

Further Math, Physics, Computer Science (GPA: 4.98 / 5.00)

May 2025

- 'Orken' Presidential Full-Ride Merit-Based Scholarship.
- ISHR (Germany) merit exchange (top 15 out of 400 students).
- Johns Hopkins Center for Talented Youth – Mathematical Modeling (Game of Life).

PROJECTS

COVID-19 Epidemiological Simulation | *Python, NumPy, Matplotlib*

- Built a computational SIR model to simulate COVID-19 transmission dynamics; implemented Euler's method numerical solver from scratch, balancing step-size efficiency with mathematical accuracy across 5 epidemic scenarios.

Predictive Analysis of Breast Cancer Diagnostics | *Python, Pandas, SciPy, Matplotlib*

- Performed statistical inference and linear regression on the Breast Cancer Wisconsin dataset (569 samples); engineered custom functions to compute descriptive statistics, ran independent two-sample t-tests, and calculated Cohen's d effect sizes.
- Developed a regression model predicting tumor concavity from cell radius; validated assumptions with Residuals vs. Fitted and Normal Q–Q plots, achieving statistically significant fit ($p < 0.05$).

SKILLS

- **Languages:** Python, C/C++, SQL, JavaScript, HTML/CSS
- **Frameworks, Libraries, Tools:** Git, Node.js, React, Three.js, React Three Fiber, WebGL, NumPy, Pandas, SciPy, Matplotlib
- **Human Languages:** English (Fluent), Kazakh (Native), Russian (Native), German (Beginner)